Aditya Dendukuri

PhD Candidate
University of California, Santa Barbara

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Research Interests

My research is in numerical linear algebra and scientific computing, including sparse and structured direct solvers, factorization reuse, Krylov methods and matrix functions, and randomized preconditioners, with applications to the simulation of and inference for large Markov chains and stochastic reaction networks. I develop these methods as both analysis and open-source C++ and Julia software.

Education

University of California, Santa Barbara CA Sept 2021 — Present
PhD in Computer Science
Advisor: Linda R. Petzold
Coursework: Matrix Analysis and Computation, Numerical Simulation, Finite Element Methods, Level Set Methods, Spectral Graph Theory

University of Arkansas, Fayetteville AR Aug 2018 — Jul 2020
MS in Computer Science
Advisors: David M. Ford and Matthew J. Patitz
Thesis: Nonlinear Dimensionality Reduction for the Thermodynamics of Small Clusters of Particles

University of Central Arkansas, Conway AR Aug 2014 — May 2018
BS in Computer Science (Minor in Physics)
Advisor: Tansel Halic

Research and Engineering Experience

Lawrence Berkeley National Laboratory Berkeley, CA
UC Graduate Student Researcher (GSR) Intern July 2026 – Present
Applied Mathematics and Computational Research Division

- Develop algorithms that accelerate simulation of large continuous-time Markov chains by eliminating sparse subnetworks through Schur complements and first-exit distributions.
- Implemented block-tridiagonal LU and block Cholesky factorizations that refactor only from the first changed block, reusing earlier factorizations across subnetworks.
- Use Woodbury updates of retained inverses, and AC2 approximate Cholesky as the preconditioner for randomized inverse-diagonal estimates that select states to eliminate.
- Develop randomized algorithms that sample directly from triangular factors of structured M-matrices.

University of California, Santa Barbara Santa Barbara, CA
Graduate Research Assistant, advised by Linda R. Petzold Sept 2021 – Present

- Flux-preserving adaptive finite state projection for multiscale stochastic reaction networks (accepted, *SIAM Multiscale Modeling & Simulation*).
- Multigrid finite state projection for spatially resolved reaction networks, in which coarse models guide refinement (Copper Mountain Conference on Multigrid Methods, 2026).
- Adjoint Krylov gradients for parameter inference in affine families of linear evolution equations, sharing Krylov bases across large operator libraries (submitted, *SISC*).
- Koopman operator inference of reaction rates without forming the chemical master equation (submitted, *SIAM Journal on Applied Dynamical Systems*).
- Time-series and reduced-order analysis of firing sequences in human brain organoids (*Nature Neuroscience*, 2026).

NSF National Center for Atmospheric Research (NCAR) Boulder, CO
SIParCS Graduate Student Intern May 2024 – July 2024

- Ported parts of the TUV-x radiative transfer code from Fortran to modern C++.
- Implemented the C++ tridiagonal (Thomas) solver used by the Delta-Eddington solver and benchmarked it against LAPACKE; added Google Test and Google Benchmark coverage.

University of Arkansas Fayetteville, AR
Research Associate Sept 2020 – June 2021

- Wrote performant pure-C code for Monte Carlo nucleation studies and critical-cluster analysis.

University of Arkansas Fayetteville, AR
Graduate Research Assistant Aug 2018 – July 2020

- Reconstructed energy landscapes and transition pathways of small particle clusters from stochastic simulations with data-driven methods (*Computers & Chemical Engineering*).
- Simulated quantum circuits to study hybrid quantum-classical learning models.

University of Central Arkansas Conway, AR
Undergraduate Research Assistant Aug 2016 – May 2018

- Real-time surgical training simulation coupling fluid-flow and heat-transfer models (IEEE BIBE).

Preprints (Under Review)

- A. Dendukuri, S. Chandrasekaran, and L. R. Petzold, **An adjoint Krylov approach to parameter inference for affine families of linear evolution equations**, Submitted to *SIAM Journal on Scientific Computing*, 2026. arXiv: 2610.15940 [cs.CE]. [Online]. Available: https://arxiv.org/abs/2610.15940
- A. Dendukuri and L. R. Petzold, **Koopman operator inference for the chemical master equation**, Submitted to *SIAM Journal on Applied Dynamical Systems*, 2026. arXiv: 2610.15937 [cs.CE]. [Online]. Available: https://arxiv.org/abs/2610.15937

Publications

- A. Dendukuri, S. Chandrasekaran, and L. Petzold, **Flux-preserving adaptive finite state projection for multiscale stochastic reaction networks**, 2025. arXiv: 2512.17064 [cs.CE]. [Online]. Available: https://arxiv.org/abs/2512.17064 Accepted for publication in *SIAM Multiscale Modeling & Simulation* (in press).
- T. van der Molen, A. Spaeth, M. Chini, *et al.*, **Preconfigured neuronal firing sequences in human brain organoids**, *Nature Neuroscience*, vol. 29, no. 1, pp. 123–135, Jan. 2026, ISSN: 1546-1726. DOI: 10.1038/s41593-025-02111-0. [Online]. Available: https://doi.org/10.1038/s41593-025-02111-0
- A. Dendukuri, M. Tunc, D. Demirel, *et al.*, **Unified framework for real-time fluid simulation in virtual rotator cuff arthroscopic skill trainer (vircast)**, in *2023 IEEE 23rd International Conference on Bioinformatics and Bioengineering (BIBE)*, 2023, pp. 346–353. DOI: 10.1109/BIBE60311.2023.00063
- D. M. Ford, A. Dendukuri, G. Kalyoncu, *et al.*, **Machine learning to identify variables in thermodynamically small systems**, *Computers & Chemical Engineering*, vol. 141, p. 106 989, 2020, ISSN: 0098-1354. DOI: https://doi.org/10.1016/j.compchemeng.2020.106989. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0098135419311937
- D. Demirel, A. Yu, S. Cooper-Baer, *et al.*, **A hierarchical task analysis of shoulder arthroscopy for a virtual arthroscopic tear diagnosis and evaluation platform (vatdep)**, *The International Journal of Medical Robotics and Computer Assisted Surgery*, vol. 13, no. 3, e1799, 2017

Software

- **numerics** – C++20 numerical linear algebra library, developed solely by me, with dense, banded and sparse factorizations, Krylov solvers, matrix exponentials, Woodbury factorization reuse and mixed-precision LU. It has optional BLAS, LAPACK, OpenMP, SuiteSparse and FFTW back ends, hand-written CUDA kernels, C++20 concepts for operator properties, and Google Test and Google Benchmark suites. Documentation, GitHub.
- **markovkit** – C++20 library for continuous-time Markov chains with adaptive and Krylov finite state projection, stochastic simulation, multigrid-preconditioned reaction-diffusion solvers, and subnetwork elimination. Documentation, GitHub.
- **AdjointKrylovGradients** – Julia implementation and reproducibility code for the adjoint Krylov gradient paper. GitHub.
- **approxchol** – Header-only C++20 randomized approximate Cholesky for graph Laplacians, under 200 lines. GitHub.

Conference Presentations

- **Data-Driven Rate Recovery and Sensitivity Analysis for Discrete Stochastic Chemical Reaction Systems.** SIAM Annual Meeting, Cleveland, OH, 2026.
- A. Dendukuri and L. Petzold, **Adaptive multigrid finite state projection for the reaction-diffusion master equation**, Colorado Conference for Iterative and Multigrid Methods, Boulder, CO, USA, June 26th, 2026

- A. Dendukuri, M. Tunc, D. Demirel, *et al.*, **Unified framework for real-time fluid simulation in virtual rotator cuff arthroscopic skill trainer (vircast)**, 23rd IEEE International Conference on BioInformatics and BioEngineering (BiBE 2023). Presented by Doga Demirel, Oct 20-24, 2019
- A. Dendukuri, G. Kalyoncy, K. Luu, *et al.*, **Detecting subspaces for small lennard-jones clusters using spectral graph theory and machine learning**, 2021 SIAM Conference on Computational Science and Engineering, ., March 20-21, 2021
- D. M. Ford, A. Dendukuri, G. Kalyoncu, *et al.*, **Machine learning to identify variables in thermodynamically small systems**, Foundations of Process Analytics and Machine Learning., Aug 6-9, 2019
- A. Dendukuri, B. Keeling, A. Fereidouni, *et al.*, **Defining quantum neural networks via quantum time evolution**, Quantum Techniques in Machine Learning 2019, KAIST, Daejon, South Korea. ., Oct 20-24, 2019
- A. Dendukuri, T. Mustafa, T. Halic, *et al.*, **Modeling fluid and heat flow in virtual arthroscopic tear diagnosis and evaluation platform (vatdep)**, Arkansas IDeA Network of Biomedical Research Excellence symposium. Fayetteville, AR., Oct 25-26, 2017
- A. Dendukuri, T. Halic, S. Kockara, *et al.*, **Modeling fluid flow in virtual arthroscopic tear diagnosis and evaluation platform (vatdep)**, Sixth Annual Central Arkansas Undergraduate Summer Research Symposium. University of Arkansas for Medical Sciences, Little Rock, AR., Jul 26, 2017

Teaching and Service

- Graduate Teaching Assistant at UCSB (2021 – Present) for scientific computing, numerical methods, and data structures (C++, Python, MATLAB).
- Session chair, CP17: Stochastic Systems, SIAM Annual Meeting, Cleveland, OH, 2026.

Honors

- Best paper award. IEEE Bioinformatics and Bioengineering (BIBE) Conference 2023.
- Masters thesis nominated for the *Conference of Southern Graduate Schools (CSGS)* outstanding master's thesis award.
- Graduate Student Council Travel Grant ($500) for the Foundations of Process Analytics and Machine Learning Conference in Raleigh, North Carolina (Summer 2019).
- Advancement of Undergraduate Research in the Sciences (AURS) summer research fund (2016) (`http://www.aursfund.org`) Award Amount: $5000

Technical Skills

- **Programming:** C++ (C++17/20, templates, concepts), C, CUDA, Julia, Python
- **Numerical linear algebra:** sparse direct factorizations, Schur complements, factorization reuse, Krylov methods and preconditioning, randomized approximate Cholesky, matrix functions
- **High-performance computing:** OpenMP, CUDA, cache blocking, vectorization, benchmarking
- **Libraries and tools:** BLAS, LAPACK, SuiteSparse, FFTW, CMake, Google Test, Google Benchmark, Git, Linux; PyTorch, SciML (Julia)

References

Dr. Linda R. Petzold (petzold@engineering.ucsb.edu)
Mehrabian Distinguished Professor, Mechanical Engineering and Computer Science, University of California, Santa Barbara.

Jian Sun (sunjian@ucar.edu)
Soft Eng/Prog II, NSF National Center for Atmospheric Research

Dr. David Ford (ford3dm@cmich.edu)
Dean, Bagley College of Engineering, Mississippi State University.
Past: Professor and Department Head. Ralph E. Martin Department of Chemical Engineering, University of Arkansas.

Dr. Tansel Halic (tanselhalic@gmail.com)
Senior Software Engineer, Intuitive Surgical.